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Review

Timber Harvesting in Mountainous Regions: A Comprehensive Review

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Abstract: Mountain ecosystems play a crucial role in providing ecosystem services, with some of the most important being carbon sequestration, biodiversity conservation, land protection, and water source preservation. Additionally, timber harvesting in these regions presents significant environmental, economic, and social challenges. This study provides a comprehensive bibliometric and systematic analysis of publications on timber harvesting in mountainous areas, examining the current state, global trends, key contributors, and the impact of forestry operations. A total of 357 publications on timber harvesting in mountainous areas have been identified, spanning from 1983 to 2024. These publications predominantly originate from the USA, Canada, Australia, and China, with additional contributions from European institutions. The research is published in leading forestry, ecology, and environmental science journals, highlighting its global impact. This study provides an in-depth bibliometric and systematic analysis, assessing research trends, key contributors, and their influence on scientific advancements in sustainable forestry and ecological conservation. These articles belong to the scientific fields of Environmental Science and Ecology, Forestry, Zoology, and Biodiversity Conservation, among others. They have been published in numerous journals, with the most frequently cited ones being Forest Ecology and Management, Journal of Wildlife Management, and Forests. The most frequently used keywords include dynamics, management, and timber harvest. The analysis of publications on timber harvesting in mountainous areas highlights the widespread use of primary harvesting methods, the negative effects of logging activities on soil, forest regeneration processes, and wildlife populations, as well as the role of advanced technologies in improving harvesting efficiency. While sustainable management practices, such as selective cutting and low-impact harvesting techniques, can mitigate some negative effects, concerns remain regarding soil erosion, habitat alteration, and carbon emissions. This analysis underscores the need for flexible forest management strategies that balance economic efficiency with ecological sustainability. Future research should focus on innovative harvesting techniques, adaptation measures to terrain and climate conditions, and the long-term impact of forestry activities on mountain ecosystems.

Keywords: timber harvesting; forest disturbances; bibliometric review; mountain forests; biodiversity conservation

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1. Introduction

One of the main types of vegetation in the mountains is the montane forest. Mountain forests cover large areas of mountain ranges such as the Alps, Pyrenees, Balkans, and Carpathians in Europe; the Appalachian and Rocky Mountains in North America; the Australian Alps; the Guiana Highlands in South America; the mountains of Central Africa; and the Andes in South America. Additionally, other significant European mountain ranges, such as the Massif Central (France), the Jura Mountains (France and Switzerland), the Dinaric Alps (Southeastern Europe), the Scandinavian Mountains (Norway and Sweden), the Cantabrian Mountains (Spain), and the Scottish Highlands (UK), also host unique mountain forest ecosystems with their own specific forestry characteristics.

The climate in mountainous regions strongly influences the distribution of forests, as temperatures decrease with increasing altitude, determining their distribution and characteristics. The most widespread forests are found at mid-altitudes, where precipitation and a temperate climate favor forest growth and development. At higher altitudes, the climate is harsher, with lower temperatures and stronger winds, which prevent tree growth and lead to a transition from forests to mountain shrubs, alpine tundra, and meadows [1].

Montane forests occur between the submontane and subalpine zones. The elevation at which one habitat transitions into another varies globally, particularly by latitude. The upper limit of montane forests is often marked by a shift to hardier species that occur in less dense stands [2]. For example, in the Sierra Nevada of California, montane forests contain dense stands of lodgepole pine and red fir, while the Sierra Nevada subalpine zone features sparse stands of whitebark pine [3].

Mountain ecosystems provide numerous ecosystem services, the most important being carbon storage, water resource protection, protection against natural hazards, recreation, and timber production [4–6].

To maintain biodiversity and ensure the provision of multiple ecosystem services, various management activities and forestry operations (such as maintenance, silvicultural treatments, and product harvesting) are necessary. In many countries, under difficult terrain conditions, the costs of forest management and harvesting are not covered by timber revenues, and forest management that fails to cover costs threatens the long-term provision of ecosystem services [7].

Bibliometric analysis is a widely used and systematic approach for examining extensive scientific data. It helps reveal the developmental trends within a particular domain while highlighting emerging research areas. This method is valuable for interpreting and organizing accumulated scientific knowledge in established disciplines by processing large sets of unstructured information effectively. Recently, bibliometric review techniques have gained traction due to advancements such as specialized software, interdisciplinary strategies, and enhanced data-handling capabilities [8]. Since bibliometric methods are inherently quantitative, they mitigate selection bias often found in systematic reviews. Additionally, they are effective in assessing journal impact, tracking co-authorship networks, analyzing co-citation patterns, and identifying foundational research themes within a field [9].

So far, numerous review articles (including bibliometric reviews) have been published on mountain areas [10–13] and timber harvesting [14–16]. However, we have not identified any study that addresses both topics together. For this reason, and because we believe such a study would be valuable for researchers in these fields, we have undertaken this work.

This study aims to provide a comprehensive review of timber harvesting in mountainous regions, with a focus on its environmental, economic, and management implications. Through a bibliometric and systematic analysis, the study aims to identify global research trends, key contributors, and the primary challenges associated with forestry operations in high-altitude forests. The purpose of this research is to assess the impact of timber

harvesting on forest ecosystems, including soil stability, biodiversity, carbon sequestration, and water resources. Additionally, it seeks to evaluate existing harvesting methods, technological advancements, and sustainable management practices that can enhance efficiency while minimizing ecological disturbances. By synthesizing existing knowledge, this study aims to inform policymakers, researchers, and forestry professionals on best practices for balancing timber production with conservation goals in mountainous environments.

2. Materials and Methods

To compile a bibliographic database on timber harvesting in mountainous regions, we utilized the Web of Science Core Collection, covering publications from 1 January 1983 to 31 December 2024. This dataset provides insights into long-term research trends and the evolution of scientific contributions in this field. The primary source of Web of Science data and searches with the assumed keywords "logging in mountainous areas" for the aspects of types of publications, Web of Science categories, years of publication, geographic distribution, affiliated institutions, language of publication, journals, publishers, authorship, and keywords in publications are indicated.

Data analysis was performed using Web of Science Core tools [17], Microsoft Office 2016 Plus [18], and Geochart [19]. For visual mapping and cluster identification, VOSviewer (version 1.6.20) [20] was employed. Initial screening involved removing conference papers, articles with ambiguous origins, irrelevant studies, and duplicate entries. Out of an initial 427 articles, 358 remained after filtering and underwent further analysis. This bibliometric assessment aimed to highlight emerging trends, key contributors, and significant insights into research articles, authors, and journals in this domain. Ten major aspects were examined: (1) publication types, (2) Web of Science categories, (3) publication years, (4) geographical distribution, (5) affiliated institutions, (6) language of publication, (7) journals, (8) publishers, (9) authorship, and (10) keywords.

In interpreting the data, we incorporated terms such as Total Link Strength and Node Size. Total Link Strength, a concept frequently discussed in scientific literature, illustrates the extent of collaboration between researchers or the effectiveness of research groups. Evaluating an article's Total Link Strength is crucial, as two key indicators determine its impact: the Links attribute and the Total Link Strength attribute. The Links attribute reflects the number of direct connections an article shares with other publications, whereas Total Link Strength assesses the overall intensity of those connections through co-authorship networks. For instance, in co-authorship analysis, the Links attribute quantifies the number of direct co-author relationships a researcher has, while Total Link Strength measures the overall strength of those connections within the academic landscape. This methodology enables the identification of research groups working on specialized topics and evaluates the role of collaborations in influencing scientific progress.

For keyword analysis, the node size and the thickness of the connecting lines in the visual representation are directly proportional to the number of documents in which a given keyword appears. Colors indicate the respective clusters, while the connecting lines between nodes signify keyword co-occurrence. A shorter distance between nodes denotes a stronger relationship between the keywords.

In the second phase of analysis, a traditional review method was employed to conduct an in-depth examination of 358 articles. The results were classified into six primary categories: timber harvesting in mountain areas after disturbances; methods and programs used in timber harvesting in mountain areas; influence of timber harvesting in mountain areas on wildlife; influence of timber harvesting in mountain areas on soils; other influences of timber harvesting in mountain areas; timber harvesting systems in mountain areas (Figure 1).

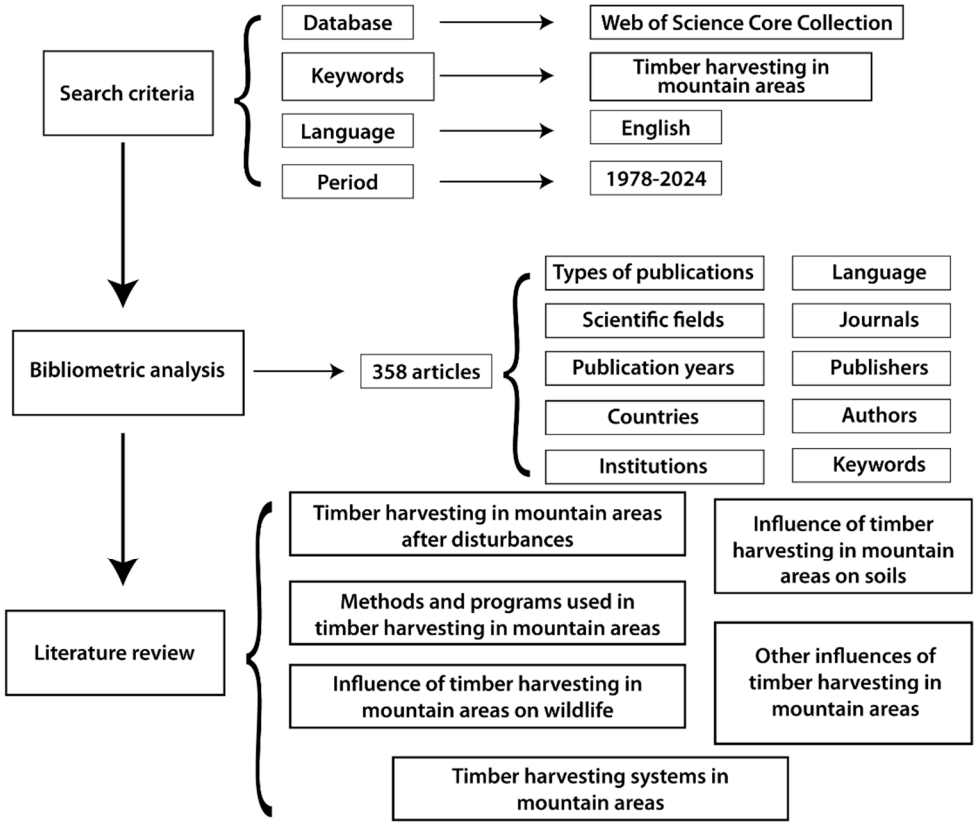


Figure 1. Used methodology.

3. Results

3.1. A Bibliometric Review

A total of 358 publications on timber harvesting in mountainous areas have been identified. The vast majority are articles (320 articles, accounting for 89% of the total publications), followed by 28 proceedings papers (8%), nine reviews (3%), and one book chapter (Figure 2).

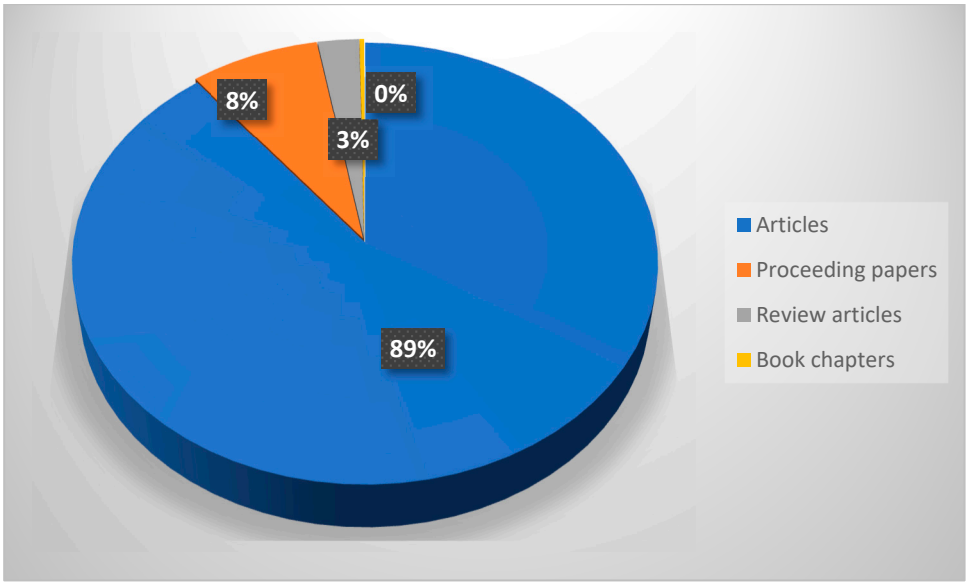


Figure 2. The composition of selected item database.

The analysis of publications related to timber harvesting in mountainous areas reveals that the most prolific author in this field is David Lindenmayer, with 10 articles, followed by Hong He (six articles), Michael Aust (five articles), and W. M. Ford (five articles). These contributions highlight significant research interest and expertise in the domain.

Institutional affiliations indicate that the majority of publications originate from institutions based in the United States. The top five institutions include the United States Department of Agriculture (USDA) with 58 articles, the United States Forest Service (55 articles), the United States Department of the Interior (31 articles), the United States Geological Survey (22 articles), and Oregon State University (16 articles). Beyond North America, notable contributions come from Australia (Australian National University, 12 articles) and China (Chinese Academy of Sciences, 11 articles). In total, over 200 institutions worldwide have contributed to this field of study.

Regarding language distribution, the vast majority of articles (335) are published in English, with smaller contributions in Polish (six articles), Russian (six articles), Czech (two articles), and single entries in Croatian, German, and Portuguese.

The leading publishers in this research domain include Elsevier (98 articles), Wiley (49 articles), Springer Nature (30 articles), MDPI (16 articles), Oxford University Press (11 articles), and Taylor & Francis (10 articles), among others, totaling over 80 different publishers.

Figure 3 presents, in the form of a histogram, the main research fields that accounted for over 84% of the published results between 1978 and 2025. The field of Environmental Science and Ecology leads with a share of over 26%, followed closely by Forestry, which represents 25.7% of the total results. At a considerably lower percentage, the fields of Zoology (7.42%), Biodiversity Conservation (6.47%), and Geology (4.19%) follow.

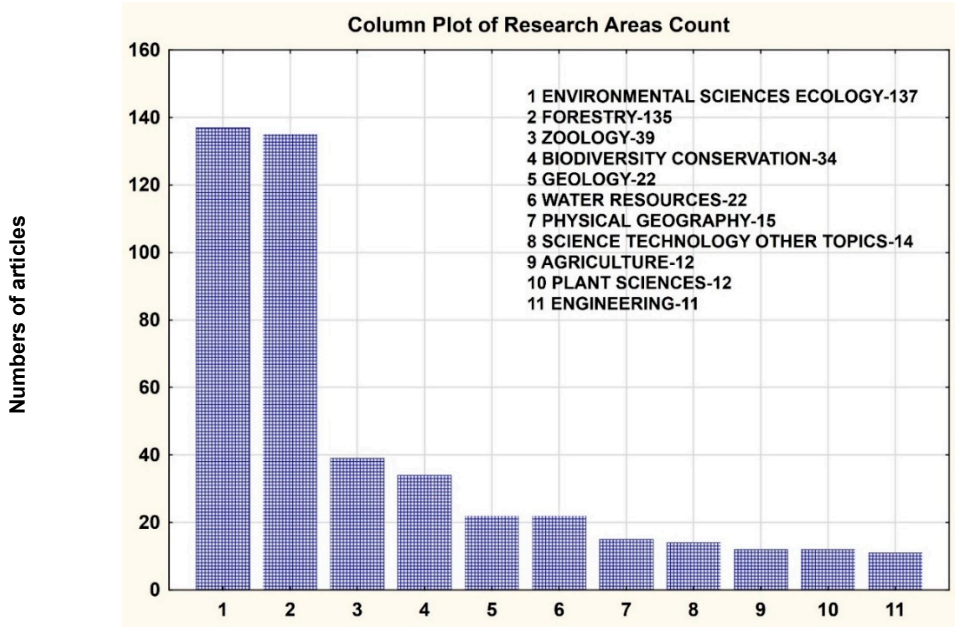


Figure 3. Histogram of the main research fields.

Figure 4 shows the distribution of published contributions by year. A unimodal distribution with an increasing trend can be observed. While between 1976 and 1995, only 26 articles were published (less than 5% of all selected articles), in the most recent period (2020–2025), scientific production has reached a much higher level. However, statistical analysis does not indicate a clear linear increasing trend ($R^2 = 0.624$), nor is there

a significant dependence of the volume of scientific production on time ($p = 0.47 > 0.05$). This suggests that interest in this field has fluctuated over time.

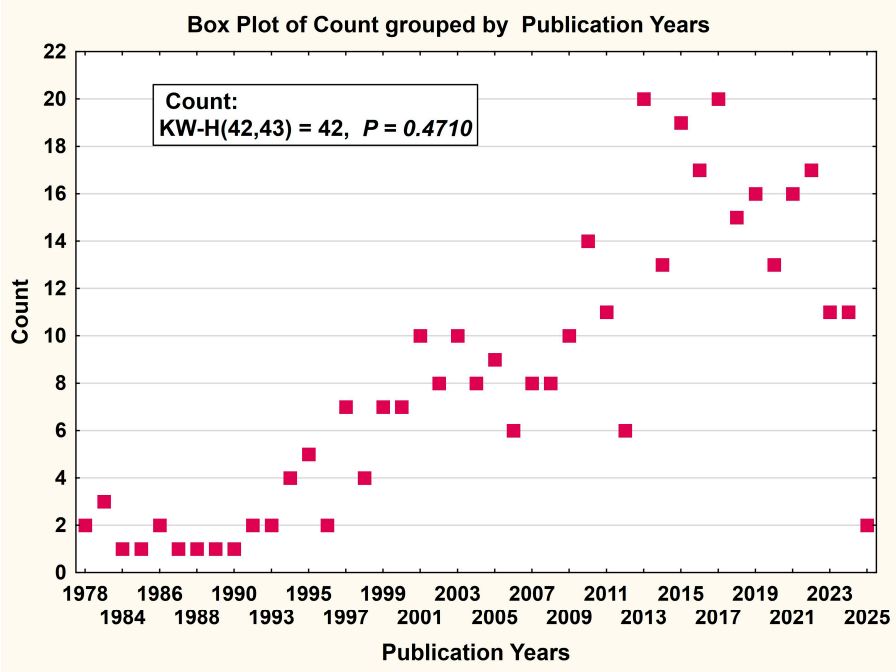


Figure 4. Representation of the number of publications by year.

Geographically, authors contributing to this research span 63 countries across five continents (Figure 5). The United States leads in terms of published articles (194), followed by Canada (51), Australia (26), and China (18). When considering research collaboration strength, the highest ranked countries are England, the USA, and Indonesia. Based on Total Link Strength, the hierarchy is USA, Germany, and Australia (Table 1).

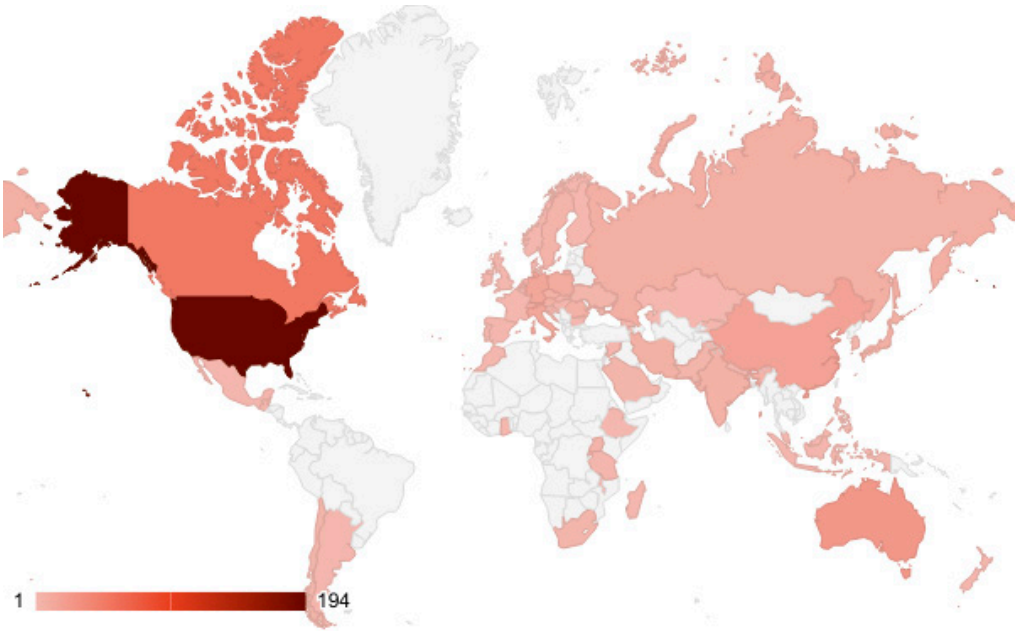


Figure 5. Countries with authors of articles on timber harvesting in mountainous regions.

Table 1. The most representative countries where articles on timber harvesting in mountainous regions have been published.

Crt. No.	Country	Documents	Citations	Total Link Strength
1	USA	183	5818	65
2	Germany	15	571	28
3	Australia	22	893	17
4	Switzerland	8	429	16
5	Canada	49	1175	15
6	Austria	7	255	14
7	China	18	266	14
8	Ukraine	3	260	13
9	Slovakia	4	170	11
10	Italy	14	277	10
11	Sweden	4	10	9
12	Iran	7	479	6
13	Japan	7	502	6
14	Poland	10	135	6

Table 1 illustrates the distribution of scientific publications across the most influential journals covering timber harvesting in mountainous areas. Additionally, the table provides an assessment of the Total Link Strength attribute, which quantifies the intensity of co-authorship connections between researchers and their collaborators.

The countries of origin of the authors who have published articles on this topic can be grouped into nine clusters, each containing at least four countries (Figure 6).

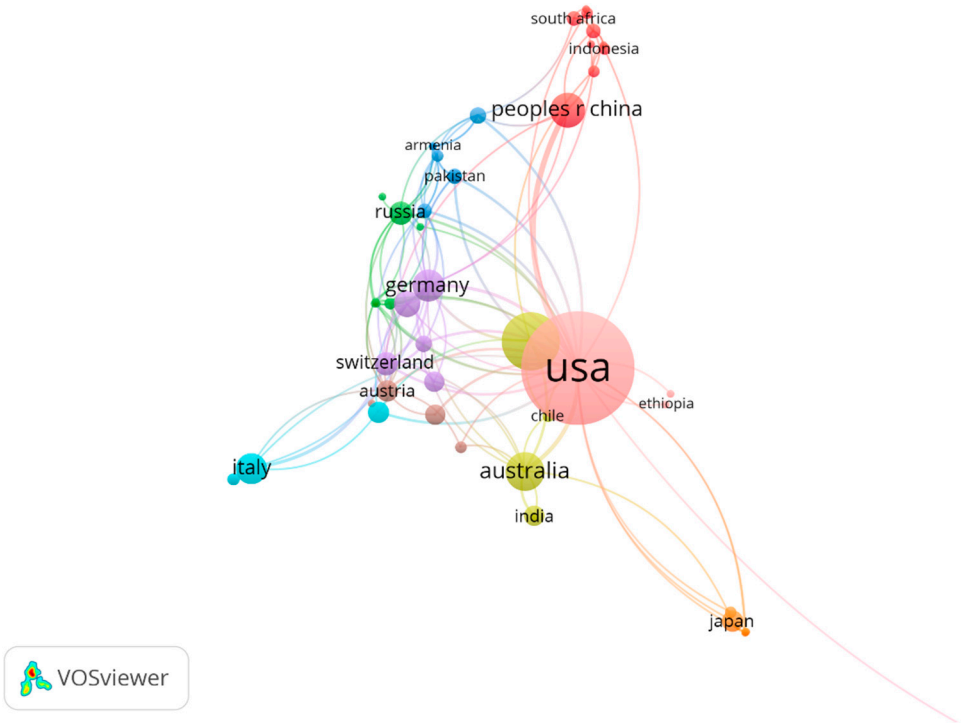


Figure 6. Clusters of countries with authors of articles on timber harvesting in mountain areas.

The top five clusters are as follows: cluster 1: Ghana, Indonesia, Netherlands, Norway, China, Rwanda, South Africa, Tanzania, and Uganda; cluster 2: Belgium, Finland, Hungary, Kazakhstan, Luxembourg, Madagascar, and Russia; cluster 3: Armenia, Georgia, Kyrgyzs-

tan, Pakistan, Saudi Arabia, Sweden, and Ukraine; cluster 4: Australia, Canada, Chile, and India; cluster 5: Czech Republic, Germany, Poland, Slovakia, and Switzerland.

The bibliometric analysis identified significant research contributions from North America (USA, Canada), Australia, and China, with additional input from European countries such as Germany, Switzerland, and Austria.

The USA leads in total publications (194 articles), with major contributions from institutions such as the United States Department of Agriculture (USDA) and the United States Forest Service. Canada follows with 51 publications, primarily from universities and forestry research organizations. Australia and China contribute 26 and 18 articles, respectively, focusing on sustainable logging practices and the impact of climate change on mountainous forests. European research is distributed across multiple countries, with Germany, Switzerland, and Austria playing significant roles in advancing forest management technologies.

According to our statistics, the journals with the highest number of published articles on this topic are as follows, based on the number of articles: *Forest Ecology and Management*, *Journal of Wildlife Management*, and *Forests*. Based on Total Link Strength, the most significant journals are *Forest Ecology and Management*, *Journal of Wildlife Management*, *Biological Conservation*, and *Journal of Hydrology* (Table 2, Figure 7).

Among the 2410 keywords, the most frequently used are dynamics, management, and timber harvest (Table 3).

Table 2. The most representative journals where articles on timber harvesting in mountainous regions have been published.

Crt. No.	Journal	Documents	Citations	Total Link Strength
1	*Forest Ecology and Management*	53	2086	30
2	*Journal of Wildlife Management*	12	418	9
3	*Biological Conservation*	9	489	7
4	*Journal of Hydrology*	5	897	7
5	*Ecological Applications*	5	347	5
6	*Hydrological Processes*	4	156	5
7	*Landscape Ecology*	7	400	5
8	*Remote Sensing of Environment*	4	374	5
9	*Wildlife Society Bulletin*	4	105	4
10	*Canadian Journal of Forest Research*	7	103	3
11	*Conservation Biology*	3	230	3
12	*Environmental Research Letters*	3	80	3
13	*Journal of Raptor Research*	4	50	3
14	*Wildlife Monographs*	3	137	3
15	*Forests*	10	144	2

With the help of the VOSviewer software, the keywords can be grouped into clusters, with the most significant being, in the first cluster: biodiversity, catchment, clear-cut, coarse woody debris, dynamics, impact, logging, mountains, patterns, responses, roads, small mammals, species richness, and timber harvest; the second cluster: biomass, climate-change, disturbance, Douglas fir, forests, forestry, forest management, growth, habitat selection, landscape, model, mountain pine beetle, timber, and vegetation; the third cluster: climate, ecosystem services, fire, impacts, management, regeneration, restoration, simulation succession, and wildfire (Figure 8).

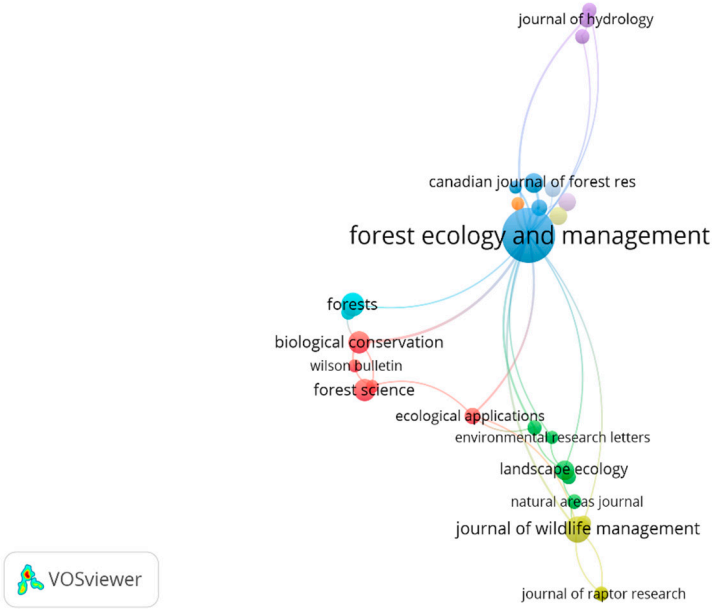


Figure 7. The primary journals publishing research on timber harvesting in mountainous regions. The node size and thickness of the connecting lines are proportional to the number of documents assigned to each country. The connections represent the collaboration network between research institutions.

Table 3. The most commonly appearing keywords in studies related to timber harvesting in mountainous regions.

Crt. No.	Keyword	Occurrences	Total Link Strength
1	dynamics	38	134
2	management	42	129
3	forest management	40	114
4	timber harvest	45	112
5	disturbance	29	91
6	fire	24	77
7	forest	29	75
8	conservation	30	73
9	patterns	24	69
10	growth	19	65
11	impacts	22	64
12	biodiversity	20	59
13	climate change	17	56
14	habitat	18	56
15	abundance	18	54
16	vegetation	15	45

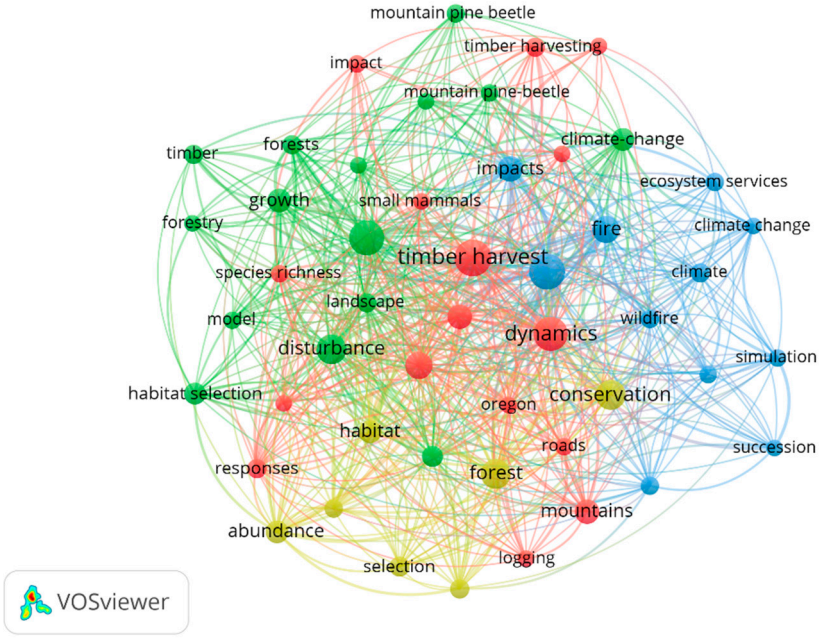


Figure 8. Authors' keywords concerning timber harvesting in mountainous regions.

The progression of keyword usage over time reveals distinct trends: between 2009 and 2012, commonly used terms included disturbance, model, selection, habitat, and abundance; from 2013 to 2015, the focus shifted to fire, impacts, landscape, management, and dynamics; whereas in the 2015–2017 period, keywords such as climate change, ecosystem services, wildfire, and mountain pine beetle gained prominence (Figure 9).

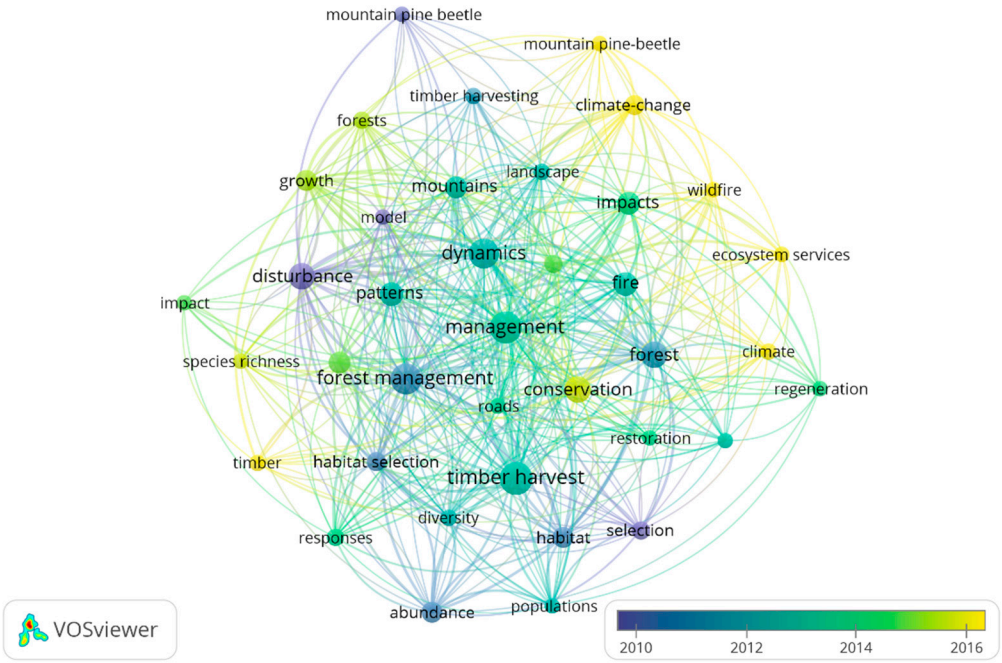


Figure 9. Annual distribution of keywords concerning timber harvesting in mountainous regions.

3.2. Literature Review

3.2.1. Timber Harvesting in Mountain Areas After Disturbances

Following catastrophic disturbances, forest managers often implement salvage harvesting to recover economic losses. However, when disturbances occur unpredictably, an inherent option value emerges, representing the benefit of delaying salvage operations

to gather more information about forest recovery. Analytical studies suggest that slower forest recovery rates justify postponing salvage harvesting, whereas areas with high timber value or where non-timber values are highly sensitive to dead and dying trees should be harvested immediately [21].

Forest disturbances significantly shape mountain ecosystems, influencing forest regeneration, biodiversity, and carbon dynamics. Timber harvesting is often employed following such disturbances to recover economic value, mitigate risks, and support ecological recovery. These disturbances can be categorized as biotic and abiotic disasters, each with distinct impacts and management considerations.

Biotic Disasters

Beetles

Salvage harvesting of mountain pine beetle-infested forests in the northern Colorado Rocky Mountains offers economic and environmental benefits by utilizing otherwise wasted resources, generating revenue, and reducing greenhouse gas emissions. While traditional timber and bioenergy production are often managed separately, integrating both can enhance resource efficiency. Although the market value of beetle-killed wood is relatively low and harvesting costs are high, salvage operations can still yield beneficial greenhouse gas emission reductions [22].

In the southern Rocky Mountains, beetle outbreaks have transformed millions of hectares of lodgepole pine forests, setting them on new development trajectories. A study by Collins et al. [23] found that new seedling colonization occurred in both harvested and untreated stands, but seedling density—primarily lodgepole pine and aspen—was four times higher in harvested areas.

Salvage harvesting is commonly used to remove disturbance-killed trees after bark beetle outbreaks, aiming to recover economic value, reduce the risk of further disturbances, and support ecosystem recovery [24,25]. While previous research has extensively examined salvage harvesting after fire events [26,27] and, to a lesser extent, windstorms [28,29], studies specifically addressing salvage harvesting in beetle-killed forests remain relatively scarce [30].

Abiotic Disasters

Fires

Timber harvesting significantly influences fire regimes and landscape patterns, often deviating from natural disturbance dynamics. A study in China revealed that increased timber harvesting intensity led to reduced fine fuel loads and increased coarse fuel loads. While the potential burn area varied across different harvesting methods, fire frequency remained largely unchanged. However, timber harvesting generally increased fire risk. Notably, clear-cutting reduced the mean patch size and aggregation of larch forests while increasing these metrics for white birch forests. As a result, selective-cutting methods are recommended for sustainable forest management [31].

Further research on post-fire vegetation recovery in areas with a history of fire suppression and timber harvesting indicated that these factors contributed to denser shrub regeneration but limited sapling recovery. This suggests that previous management actions significantly influence post-fire forest regeneration [32].

Avalanches

Clear-cut logging in mountainous terrain, such as in British Columbia, Canada, is creating new snow avalanche start zones. These zones have the potential to produce avalanches large enough to penetrate and destroy mature forest cover, highlighting the need for careful consideration of harvesting practices in avalanche-prone areas [33].

3.2.2. Methods and Programs Used in Timber Harvesting in Mountain Areas

Non-mechanized and obsolete harvesting systems reported the lowest efficiency and highest environmental footprint, whereas fully mechanized systems demonstrated the highest efficiency, fewer accidents, and minimal stand damage. Cable yarders remain the preferred extraction technology for steep terrains, but they require a well-developed road network. Increasing mechanization, improving road networks, knowledge transfer, and training forest workers are crucial for enhancing timber production efficiency in European mountain forests [34].

Residue management after timber harvesting is typically conducted to mitigate fire hazards and create planting spots for seedlings, particularly on the eastern slopes of the Cascade Mountains in the Pacific Northwest. Traditionally, residue burning has been the standard practice, but air pollution concerns have prompted alternative approaches. Additionally, increasing awareness of wildlife habitat and nutrient retention has led to a shift towards leaving woody debris untreated on-site. However, the effects of these practices on seedlings and soil in drier mixed conifer forests remain understudied [35].

Tree selection for extraction in mountainous areas presents a significant challenge due to the necessity of balancing forest management plans with operational feasibility. The use of cable yarding systems in steep terrain adds complexity and costs to harvesting operations. Furthermore, mountain forests serve a protective function against natural hazards, requiring decision-makers to consider both short-term and long-term impacts while optimizing economic efficiency [36].

In South Italy's Basilicata region, ArcGIS software 10.1 was used to plan timber harvesting in five mountainous broadleaf forest areas. Factors such as slope, watershed characteristics, and natural obstacles were analyzed to identify accessible forest areas, plan additional roads for optimized timber extraction, and determine site-specific harvesting methods and cutting intensities [37].

A density-dependent matrix model (DM2) was developed in Korea for forest stands with varying conditions under different harvesting schemes. After validation with field data, the model proved effective in optimizing harvest strategies, taking into account stand growth, total timber yield, and size diversity over time [38].

An interactive web-based 3D planning tool has been created to support the entire wood processing chain in mountainous regions. This tool reconstructs virtual forest environments with high accuracy, assisting stakeholders from tree marking to timber production in sawmills. It offers novel simulation, planning, and monitoring capabilities, accommodating the needs of all involved actors [39].

To enhance cost-effectiveness in forestry, Switzerland has developed a spatial decision support system that allocates the most suitable harvesting methods to specific plots. This system considers hauling route limitations, extraction properties, and stand characteristics. The approach is based on productivity models and expert-defined decision trees [40].

The SLOPE program integrates geospatial information by combining traditional laser scanning surveys with UAV-based aerial surveys. This approach enhances accuracy in timber harvesting planning and monitoring [41].

3.2.3. Influence of Timber Harvesting in Mountain Areas on Wildlife

The identified articles related to the researched aspect are presented in Table 4.

Table 4. Animal species affected by timber harvesting in mountain areas identified in published articles on this topic.

Cur. no.	Species	Geographical Area	Cited by
	Invertebrates		
1	benthic invertebrate	USA	Hernandez et al., 2005 [42]
	Vertebrates		
	Fish		
2	salmon	Canada	Baldigo et al., 2005 [43]
	Amphibians		
3	amphibians	USA	Harper et al., 2015; Barry et al., 2008; Biek et al., 2002 [44–46]
	Reptiles		
4	salamander	USA	Mahoney et al., 2016; Knapp et al., 2003 [47,48]
	Birds		
5	birds	Argentina	Zurita and Zuleta, 2009 [49]
6	songbird species	USA	Weakland et al., 2002 [50]
7	boreal owls (*Aegolius funereus*)	Canada	Hayward, 1997 [51]
8	northern spotted owl (*Strix occidentalis caurina*)	USA	Clark et al., 2013 [52]
	Mammals		
9	small mammals	USA	Perry and Thill, 2005; Horn et al., 1983 [53,54]
10	bat	USA	Owen et al., 2004 [55]
11	Allegheny woodrat (*Neotoma magister*)	USA	Castleberry et al., 2001 [56]
12	red squirrel (*Tamiasciurus hudsonicus*)	USA	Johnson et al., 2015 [57]
13	yellow-bellied glider (*Petaurus australis*)	Australia	Incoll et al., 2001 [58]
14	caribou	Canada	Nobert et al., 2020 [59]
15	red deer (*Cervus elaphus* L.)	Poland	Bobek et al., 1984 [60]
16	reindeer	Sweden	Parks et al., 2002 [61]
17	bighorn sheep (*Ovis canadensis*)	USA	Smith et al., 1999 [62]
18	Leadbeater's possum (*Gymnobelideus leadbeateri*).	Australia	Attiwill, 1995 [63]
19	greater glider (*Petauroides volans*)	Australia	McLean et al., 2018 [64]
20	wolf (*Canis lupus*)	Canada	Kuzyk et al., 2004 [65]
21	bobcat (*Lynx rufus*)	USA	McNitt et al., 2020 [66]
22	black bear (*Ursus americanus*)	USA	Mitchell and Powell, 2003 [67]
23	grizzly bear	Canada	Ciarniello et al., 2005 [68]

3.2.4. Influence of Timber Harvesting in Mountain Areas on Soils

Timber harvesting in mountain areas influences most of the physical, chemical, or biological characteristics of soils or soil solutions, as observed in the articles extracted from the specialized literature, summarized in Table 5.

Table 5. Soil characteristics influenced by timber harvesting in mountain areas.

Cur. No.	Soil Characteristics	Specific Conditions	Cited by
1	bulk density	20-year period after timber harvesting	Ezatti et al., 2012 [69]
2	litterfall	eucalyptus plantations cut	Santiago et al., 2011 [70]
3	microbial soil composition	beech forests	Dinca et al., 2021 [71]
4	nitrogen in soil	soils in Bavarian Alps	Christophel et al., 2015 [72]
5	porosity	tilling compacted forest soils	McNabb et al., 2001 [73]
6	soil erosion	topographically different regions	Fielding et al., 2022 [74]
7	soil moisture content	landslides	Satgada et al., 2023 [75]
8	soil organic carbon	montane ash forests in Australia	Keith et al., 2014 [76]
9	soil rutting	effects of forwarder tire pressure	Eliasson, 2005 [77]
10	soil solution	water flux data for precipitation and stream discharge	Jewett et al., 1995 [78]
11	soil structure	general	Froehlich et al., 1985 [79]
12	water-holding capacity	modern forestry vehicles	Horn et al., 2004 [80]

3.2.5. Additional Effects of Timber Harvesting in Mountain Areas

Watersheds

Timber harvesting represents one of the most significant human-driven disturbances in forested watersheds, with cleared areas producing between one to five times more erosion than undisturbed land [81]. When sediments from these areas enter streams, they can degrade water quality and aquatic ecosystems [82].

Paired catchment studies have been utilized to evaluate how vegetation removal, including logging, influences streamflow patterns, particularly low and peak flows, as well as total annual water yield [83].

Streams

Research on timber harvesting and drainage basins has shown that variations in physical stream features play a crucial role in shaping the impact of logging within a given basin [84]. Large woody debris affects channel morphology, sediment transport, and overall stream geometry [85,86]. These changes occur through alterations in water flow at both the channel-unit scale [87,88] and across broader reach levels [89].

Soil erosion

Timber harvesting without adequate forestry best management practices (BMPs) can intensify soil degradation, potentially leading to reduced water quality and site productivity [74,90]. Studies indicate that infrastructure elements such as roads, skid trails, and stream crossings often expose bare soils, making them primary contributors to erosion [91–93].

Net carbon emissions

Accurate assessments of long-term carbon balance in forest harvesting require a full life-cycle analysis (LCA) framework. This includes evaluating changes in forest carbon stocks, transportation and processing of wood, carbon storage in wood-based products, and emissions reductions from substituting wood residues for fossil fuels or energy-intensive construction materials. Evaluating shifts in native forest carbon stocks necessitates examining biomass accumulation over time, soil carbon dynamics, and wildfire impacts. Research in Australia suggests that managing native forests sustainably while utilizing biomass for wood products or energy can contribute to lowering national net carbon emissions [94].

Microhabitat

While timber harvesting has been recognized as an effective tool for enhancing pollinator habitats, the role of microhabitats such as log landings remains underexplored. These open spaces, used for loading harvested logs, may provide key habitats in early successional stands due to their open conditions and high floral abundance. Studies indicate that log landings contain roughly 14 times more floral resources than harvested interiors, supporting greater numbers of bees and butterflies. Findings suggest these areas function as essential resource hubs for pollinators [95].

Landscape structure

Examining shifts in landscape patterns due to roads and logging revealed limited overall impacts when considering entire landscapes. However, on lands designated for timber production, significant changes in forest fragmentation and structure were observed. Among the factors analyzed, roads exerted a greater influence on landscape alterations than logging activities [96].

Forest

Timber harvesting in mountain environments can also lead to damage affecting residual trees, seedlings, and soil stability in these ecosystems [97,98].

3.2.6. Timber Harvesting Systems in Mountain Areas

Timber harvesting in mountainous areas presents significant challenges due to steep slopes, rough terrain, and accessibility constraints. The analysis of forestry operations worldwide indicates that mechanized logging is increasingly preferred in industrialized nations, whereas manual labor remains dominant in parts of the Asia-Pacific region [99]. Modern timber extraction technologies, such as cable yarding, skidders, and self-propelled winches, have enhanced efficiency on steep slopes. Studies show that skidding is the most commonly used extraction method (75%), followed by cable yarding (15%) and forwarding (8%). However, the efficiency of mechanized logging is dependent on factors such as road network quality, operator expertise, and the physical characteristics of the harvested trees [34].

Challenges in Mountainous Timber Harvesting:

Increased power requirements for machinery movement on slopes; stability issues during logging and transportation; variation in soil types, including rocky and moisture-sensitive soils; influence of weather conditions, such as heavy rainfall and snowmelt, which can halt operations due to erosion risks; limitations imposed by tree volume and silvicultural requirements, sometimes preventing harvester use; shortages of skilled machine operators, impacting mechanized logging efficiency [100].

The NEWFOR project has identified seven different technologies used in alpine conditions, ranging from basic chainsaw operations to advanced helicopter logging. The choice of logging system is heavily influenced by the density and quality of road infrastructure. In regions with poor road access, transporting felled timber to the nearest accessible landing site remains a significant bottleneck [101].

Regional Insights:

Cable yarding has a strong tradition in Central Europe, the Pacific Northwest (USA and Canada), and Japan, where technological advancements have led to more compact and efficient yarding systems [102]. The Appalachian region of the USA continues to rely on cable skidders, with extraction efficiency influenced by payload size and skidding distance [103].

In Southern Italy, chainsaws combined with farm tractors or animal labor remain common, though studies suggest transitioning to purpose-built harvesting systems could increase efficiency and southern Italian harvesting operations should invest in purpose-

built harvesting systems [104]. In Central Europe and in particular in the Alps, strongly related to the adoption of the cable-based harvesting system, are a large number of logging companies working with cable-based systems and in particular with four different standing skyline yarding technologies [105]. In Romania and Austria, skidders dominate in less challenging areas, whereas cable yarding is preferred for more difficult and sensitive sites [106,107]. In Poland, grand changes have also taken place in the field of skidding and timber transportation, shifting from horse-drawn carts to farm tractors and, recently, using specialist forwarders, equipped with cranes and hydraulic grabbers [108,109].

A new method for harvesting in mountain areas is helicopter logging. In recent years, interest in helicopter logging has increased in the Italian Alps, especially in the Central Alps in Italy [110], Switzerland [111], and Austria [112]

4. Discussion

4.1. Existing Literature on Timber Harvesting in Mountain Areas

We identified only two review articles published on timber harvesting in mountain areas, both from Australia: a review of the impacts of sustainable harvesting, non-harvest management, and wildfires on net carbon emissions from Australian native forests [94] and a review on managing stand structure as part of ecologically sustainable forest management in Australian mountain ash forests [113].

If the dominant number of published articles is normal, what stands out in this case is the high number of proceedings papers (8% of the total number of publications), which is due to the large number of symposiums or other specialized meetings organized on this topic.

The most prolific authors on this topic, ranked by the highest number of published articles, come from Australia, China, and the USA. Similarly, the countries with the highest number of authors have an almost identical structure: the USA, Canada, Australia, and China. The predominance of these regions is attributed to their large land areas (USA—fourth largest globally, Canada—second, Australia—sixth, China—third) as well as the significant proportion of forested land.

This analysis underscores the pivotal role of scientific collaboration in advancing forestry research. Publications from North America and Europe have significantly shaped sustainable forest management policies, influencing climate adaptation strategies, biodiversity conservation, and ecological modeling approaches. Additionally, research from China and Australia has contributed to advancements in remote sensing applications, forest inventory management, and the integration of AI-driven analytics in forestry. The strong international research networks and citation linkages indicate a growing interdisciplinary approach, merging environmental science, technology, and policymaking to improve timber harvesting practices while mitigating environmental impact.

The top three publishers of articles on this topic are the well-known publishing houses Elsevier, Wiley, and Springer Nature. They are followed by MDPI, a more recent publishing house that is steadily catching up with the top three.

The presence of journals in the environmental field (such as *Environmental Science and Ecology, Forests, Biodiversity Conservation*) at the top is expected, followed by journals related to fields affected by timber harvesting in mountain areas (*Remote Sensing of Environment, Wildlife Society Bulletin, Hydrological Processes*). However, the absence of journals specializing in timber exploitation from the top 15 journals is surprising.

In general, the most frequently used keywords are dynamics, management, and timber harvest. More interesting, however, is the evolution of keyword usage over time: between 2009 and 2012, commonly used terms were directly related to the subject; from 2013 to 2015, terms related to fields affected by this activity were more frequently used; during

the 2015–2017 period, predominant keywords were associated with highly publicized and researched phenomena, such as climate change and ecosystem services.

4.2. Disturbances and Timber Harvesting in Mountain Areas

Forest disturbances, whether natural or anthropogenic, play a crucial role in shaping mountain forest ecosystems. Climate-induced disturbances such as wildfires, storms, and bark beetle outbreaks have increased in frequency and severity, prompting widespread salvage logging. Salvage harvesting is often conducted to recover economically valuable timber, reduce perceived risk of subsequent disturbance, and enhance the recovery of disturbed areas [24,25]. However, its ecological implications remain a subject of debate.

Timber harvesting can significantly influence fire regimes, altering fuel loads and landscape patterns. Studies indicate that intensive harvesting reduces fine fuel loads while increasing coarse fuel loads, potentially heightening fire risks [31]. Fire suppression and harvesting history further impact post-fire vegetation recovery, with increased shrub regeneration and limited sapling recruitment in fire-prone forests [32]. While clear-cutting may reduce fire spread by fragmenting continuous fuel sources, selective logging is generally recommended for sustainable fire management.

Bark beetle outbreaks have devastated millions of hectares of lodgepole pine forests in North America and Europe. Salvage harvesting following beetle infestations has been implemented to mitigate economic losses and reduce greenhouse gas emissions from decaying wood [22]. However, research suggests that beetle-killed stands experience natural seedling recruitment, with harvested areas exhibiting up to four times greater seedling density [23]. This highlights the need to balance economic gains with ecosystem regeneration potential when planning salvage operations.

Windstorms are another major disturbance in mountain forests, leading to large-scale tree mortality. Salvage logging following windthrow events aims to recover marketable timber and reduce fuel loads that could exacerbate future disturbances. However, studies indicate that extensive removal of fallen trees disrupts natural nutrient cycling and habitat formation [28,29]. In some cases, leaving coarse woody debris in situ supports biodiversity by providing habitat for fungi, insects, and small mammals, crucial for maintaining ecological stability.

Mountain forests play a vital role in stabilizing slopes and preventing landslides. Timber harvesting, particularly clear-cutting, can exacerbate soil instability, increasing landslide risks. Research highlights that landslide frequency is significantly higher in harvested areas compared to undisturbed forests [75]. Sustainable forest management practices, such as retaining buffer strips and employing low-impact harvesting techniques, are essential to mitigate these risks.

Given the complex interactions between disturbances and timber harvesting, an adaptive management approach is necessary. This includes implementing selective logging techniques to maintain ecosystem integrity; prioritizing post-disturbance assessments to determine optimal harvesting strategies; enhancing monitoring programs to track long-term forest recovery and biodiversity impacts; integrating climate adaptation strategies to improve forest resilience against future disturbances.

In conclusion, while timber harvesting in mountain areas following disturbances can provide economic benefits, it must be carefully managed to minimize ecological consequences. A balanced approach incorporating scientific research, sustainable forestry practices, and conservation principles is crucial for maintaining the long-term health and functionality of mountain forest ecosystems.

4.3. The Role of Advanced Technologies on Timber Harvesting in Mountain Areas

A possible option for increasing the profitability of the forestry sector is the application of state-of-the-art harvesting and extraction techniques, specifically the most suitable harvesting methods [40]. The importance of improving timber harvesting efficiency and sustainable forest management strategies for multifunctional forests has also been highlighted by other authors [114], stating that such practices (by replacing fossil fuels) can retain significant amounts of carbon and that carbon sequestration through forest management is an effective way to reduce atmospheric CO_2.

The best suitable timber harvesting methods have been evaluated in several studies. Enache et al. [34] analyzed current forestry exploitation practices in European mountain forests. They highlighted that non-mechanized harvesting systems have the lowest efficiency and the highest environmental footprint, while the use of mechanized systems increases efficiency and reduces the number of accidents and instances of tree stand damage. Cable systems (funiculars or aerial systems) are considered the most suitable timber harvesting technology in steep slope areas, but it is noted that a well-developed road network is necessary to transport machinery into the forest.

Based on conducted studies, it can be concluded that through the consistent use of the most suitable timber harvesting methods or the improvement of technology, timber harvesting could become more profitable compared to current practices [40]. However, various ecological, political, social, and economic factors influence the choice of harvesting methods and systems [115].

Recently, modern methods or software programs have been used, incorporating data or results regarding timber harvesting (Forest landscape model LANDIS PRO) [116], using LANDSAT data [117], and applying machine learning techniques and GIS [118,119].

Forest wood harvesting in mountainous areas requires thorough and precise planning to prevent potential failures and challenges arising from the complex terrain. Factors such as steep slopes, limited accessibility, and the reliance on manual labor significantly impact costs but are not always fully considered at every stage due to the varying expertise and tools used by different stakeholders. While geographic information system (GIS) planning tools have proven valuable for analyzing spatial data in such environments, their specialized nature often hinders widespread adoption, particularly in a traditionally conservative industry like forestry.

4.4. The Impact of Timber Extraction in Mountainous Regions on Biodiversity, Particularly on Wildlife Populations

Timber harvesting in mountainous regions affects various components of forest ecosystems, with wildlife populations being one of the most extensively studied aspects. The majority of research suggests that timber extraction has a negative impact on animal species, primarily through habitat alteration and fragmentation.

In the United States, forest management practices often incorporate methods that simulate natural canopy disturbances, such as prescribed burns and selective timber harvesting. Numerous studies have examined the impact of these practices on various animal species. Conversely, research in Australia has predominantly focused on marsupials due to their ecological significance and geographic uniqueness.

Certain timber harvesting practices that retain key components of mature forests distributed across the landscape may align with conservation objectives for some species. For instance, maintaining elements of mature forest has been found to be compatible with the conservation of boreal owls [51]. Additionally, research has demonstrated that the response of mammal populations to changes in forest structure due to biofuel production is complex and multifaceted. Although these harvesting methods may not drastically alter

the abundance and distribution of individual small mammal species, they can influence the prevalence of common species such as deer mice and voles [120].

Aquatic species, particularly fish such as salmon inhabiting mountain rivers, are also affected by timber harvesting. Changes in water chemistry, including shifts in acidity and increased concentrations of nitrate and inorganic monomeric aluminum, can significantly impact fish populations [43].

Habitat modification due to timber extraction poses substantial challenges for forest-dwelling species. Amphibians, which are highly sensitive to fluctuations in temperature and moisture, can be particularly vulnerable to these environmental changes. The disruption of critical microhabitats may lead to declines in amphibian populations, demonstrating the significant ecological consequences of logging activities.

However, in some cases, wildlife populations may experience short-term increases following specific silvicultural treatments. For example, clear-cutting has been observed to initially boost the abundance and diversity of small mammal communities. A review by Kirkland [121] of 21 published studies documented this phenomenon, indicating that some species may temporarily benefit from the structural changes induced by clear-cutting.

Furthermore, overstory removal techniques create early successional habitats characterized by scattered residual trees, sparse canopy cover, and a diverse understory composed of grasses, shrubs, and saplings [122,123]. These structurally complex environments provide essential habitats for a range of wildlife species, including those of conservation concern [124,125]. This indirect effect of timber harvesting demonstrates how certain management strategies can facilitate habitat diversity, ultimately supporting various wildlife species.

Overall, the impact of timber harvesting on biodiversity in mountainous regions is multifaceted. While negative effects on wildlife populations are well documented, some species may benefit from specific forest management approaches. The challenge lies in balancing timber extraction with conservation efforts to ensure the long-term sustainability of mountain ecosystems and their diverse fauna.

4.5. The Ecological Effects of Logging in Mountain Areas

Timber harvesting in mountain areas significantly impacts various soil properties, as identified in the literature review. The alterations in soil bulk density, porosity, and structure following timber extraction influence long-term soil productivity and forest regeneration. For example, increased bulk density, as observed by Ezatti et al. [69], persists for up to 20 years post-harvest, limiting root penetration and water infiltration. Similarly, reductions in soil porosity due to soil compaction [73] further exacerbate these limitations. Changes in microbial soil composition are also evident, particularly in beech forests [71], which could have cascading effects on nutrient cycling and forest succession. Moreover, alterations in soil organic carbon levels [76] suggest a potential long-term impact on carbon sequestration capacity in harvested forests. The influence of timber harvesting on soil erosion is particularly pronounced in regions with steep slopes and varied topography [74], underscoring the necessity for effective soil conservation practices.

Timber harvesting in mountain regions extends beyond soil properties, affecting watershed hydrology and stream dynamics. Studies indicate that logging significantly alters streamflow patterns, increasing peak flows and overall annual water yield due to reduced evapotranspiration [83]. Furthermore, sedimentation from harvested areas degrades water quality and disrupts aquatic ecosystems [82], raising concerns about the long-term sustainability of water resources. Large woody debris plays a critical role in maintaining stream morphology, sediment transport, and habitat complexity [85,86].

However, the removal of such debris during timber harvesting disrupts these processes, necessitating strategies to preserve in-stream habitat structures.

Soil erosion and carbon balance soil erosion following timber harvesting remain primary concerns, particularly in areas where forestry best management practices (BMPs) are not adequately implemented [90]. The construction of logging roads and skid trails exposes bare soil, contributing significantly to erosion and sedimentation in nearby water bodies [91–93]. Given these challenges, erosion control measures such as replanting vegetation and using sustainable harvesting techniques must be prioritized. Timber harvesting also influences carbon dynamics, with implications for net carbon emissions. A life-cycle analysis (LCA) approach suggests that sustainably managed forests can contribute to reducing national carbon footprints, provided that harvested biomass is efficiently utilized [94]. However, the potential loss of soil carbon stocks following timber harvesting warrants further investigation to ensure climate mitigation benefits.

The role of timber harvesting in shaping microhabitats and biodiversity is complex. While clear-cut areas can enhance pollinator habitats [95], the broader ecological consequences, including habitat fragmentation and species displacement, must be carefully assessed. The presence of log landings as resource hubs for pollinators highlights the need for integrating biodiversity conservation strategies within timber harvesting plans. At the landscape scale, roads and logging activities contribute to structural changes in forested regions [96]. While overall landscape impacts may be limited, areas designated for timber production exhibit significant fragmentation, potentially affecting ecosystem connectivity and resilience.

The ecological effects of logging in mountain areas are as follows. Logging in mountain regions leads to significant soil disturbance, which can decrease the long-term productivity of forested lands [126]. Key soil physical properties such as bulk density [79], pore size distribution or porosity [73], water-holding capacity [80], and soil rutting [77] are affected primarily due to heavy machinery use during harvesting [127]. Soil compaction from these operations reduces macropores, decreases infiltration capacity, and restricts water movement [128]. To mitigate soil disturbances, skidding should be confined to areas with gentle slopes, and alternative techniques like cable yarding should be implemented in steeper terrains [69].

Logging also depletes essential nutrients from forest ecosystems. The removal of trees extracts nutrients stored in wood and bark, while slash-burning leads to nutrient losses through volatilization and erosion. Site disturbances, such as mechanical soil preparation, can cause substantial nutrient redistribution, potentially reducing long-term soil fertility unless adequate replenishment occurs [129,130].

In tropical regions, roads and haulage tracks constructed for selective logging increase sedimentation and erosion [131]. Unchecked, these infrastructure elements can develop into gullies, leading to long-term degradation of watersheds. Additionally, while selective logging contributes to carbon emissions, it also facilitates tree regeneration and carbon recovery in some ecosystems [132].

The additional environmental effects of timber harvesting, discussed in Section 3.2.5, reinforce the multifaceted impact of logging beyond soil and carbon dynamics. The influence on watersheds is particularly concerning, as sedimentation degrades water quality and alters hydrological cycles. Stream morphology is directly affected by large woody debris removal, impacting sediment transport and aquatic habitats.

The contribution of timber harvesting to net carbon emissions underscores the necessity of sustainable forest management. The utilization of biomass for energy production or long-lived wood products can partially offset emissions, but careful assessment of soil carbon dynamics and wildfire interactions is required.

Microhabitat changes due to logging activities, such as log landings providing floral resources for pollinators, highlight the need for a balanced approach that integrates biodiversity conservation. Landscape fragmentation caused by logging roads also necessitates strategic planning to minimize disruptions to forest connectivity and ecosystem resilience.

Despite the comprehensive nature of this review, several limitations must be acknowledged. While this review highlights key trends in timber harvesting in mountainous regions, there is a lack of long-term empirical data on the ecological impacts of different harvesting methods. Many studies focus on short-term effects, leaving uncertainties about the recovery of forest ecosystems over decades. Furthermore, research on the combined effects of multiple disturbances—such as logging, climate change, and natural hazards (e.g., wildfires and landslides)—remains limited. Moreover, the economic feasibility of sustainable harvesting practices in mountain forests is not thoroughly examined. Many studies highlight environmental concerns, but fewer focus on balancing ecological sustainability with economic viability, particularly in regions where forestry operations are financially challenging due to difficult terrain and high extraction costs. Future research should address these gaps by conducting long-term ecological monitoring, incorporating interdisciplinary approaches, and integrating modern technologies into forest management strategies. Additionally, more studies should explore policy frameworks that incentivize sustainable harvesting while maintaining the economic viability of forestry operations in mountainous regions.

Timber harvesting in mountainous regions significantly influences forest dynamics, affecting both ecological and socioeconomic aspects. Research indicates that timber extraction alters forest development by impacting growing stock (standing volume) and future forest growth, thereby influencing the socioeconomic conditions in large parts of Central European mountain areas [133]. Additionally, studies have highlighted the susceptibility of advance growth—young trees established under the forest canopy—to damage during logging operations, with the extent of such damage varying substantially across different regions and harvesting practices [134]. These ecological disturbances underscore the need for sustainable forest management practices that balance timber production with the preservation of forest ecosystem integrity.

4.6. Timber Harvesting Systems in Mountain Areas

Efficiency of mechanized systems in mountain forestry: The findings highlight the dominance of cable-based extraction methods in mountain forestry due to their suitability for steep slopes. However, ground-based skidders and forwarders are becoming more common where terrain and road conditions permit. The increased mechanization of mountain logging brings benefits such as improved productivity, reduced labor intensity, and enhanced worker safety. However, it also presents challenges, including high investment costs, maintenance difficulties, and environmental concerns.

Environmental and operational constraints: Mountain logging is highly susceptible to environmental constraints, including extreme weather conditions and soil sensitivity. Heavy rainfall and snowmelt can halt operations due to erosion risks, while steep slopes require specialized extraction methods. The presence of rocky substrates or moisture-retaining soils further complicates mechanized logging. Moreover, tree size plays a significant role in determining the feasibility of harvester use; overly large trees may require alternative extraction methods.

The role of road infrastructure: A well-developed road network is crucial for efficient timber extraction. Cable yarders require road access at both the harvesting and landing sites to optimize operations. Poor road infrastructure significantly increases secondary extraction costs, as logs must be transported over greater distances to reach the nearest

accessible landing. Investments in road construction and maintenance are essential for improving overall harvesting efficiency in mountain regions.

Regional variations in harvesting practices: While mechanized logging dominates in North America, Central Europe, and parts of Asia, some regions still rely heavily on manual or semi-mechanized systems. In Southern Italy, for example, traditional logging methods persist due to economic and operational constraints. Studies suggest that transitioning to fully mechanized systems could improve efficiency and competitiveness in such regions. Helicopter logging has emerged as a viable alternative for extracting timber from remote and sensitive areas, particularly in the Alps. This method reduces ground disturbance and enables access to otherwise inaccessible locations. However, its high operational costs limit widespread adoption.

Future directions and research needs. To improve the sustainability and efficiency of mountain timber harvesting, future research should focus on developing more adaptable and environmentally friendly harvesting technologies; improving road network planning to optimize extraction logistics; exploring alternative extraction methods, such as drone-assisted logging and improved winch systems; evaluating long-term ecological impacts of different harvesting methods to ensure sustainable forest management.

In conclusion, while technological advancements have significantly improved timber harvesting efficiency in mountainous regions, operational constraints such as road access, environmental sensitivity, and workforce training remain key challenges. Addressing these issues through targeted investments and research can enhance the sustainability and productivity of mountain forestry operations.

5. Conclusions

This study provides a comprehensive bibliometric and systematic analysis of publications on timber harvesting in mountainous areas, examining the current state, global trends, key contributors, and the impact of forestry operations. The analysis indicates that, although timber harvesting remains an important economic activity, it poses significant challenges related to soil erosion, biodiversity loss, water resource alteration, and carbon sequestration.

The bibliometric analysis indicates that research on this topic has increased in recent decades, with significant contributions from the United States, Canada, Australia, and China. However, despite the wealth of literature on mountain forestry and timber harvesting separately, few studies integrate both topics holistically. The findings suggest that interdisciplinary approaches, combining forestry science, ecology, and technological innovations, are necessary to develop more sustainable harvesting practices.

While technological advancements have improved logging efficiency and environmental mitigation strategies, concerns remain regarding biodiversity loss, soil degradation, and carbon emissions. Various studies document how timber harvesting influences different wildlife species. The most commonly affected taxa include invertebrates, amphibians, reptiles, birds, and mammals. Studies also indicate that soil compaction is a significant concern in frequently logged areas, influencing long-term forest productivity; soil erosion rates in harvested areas can be 1.5 to five times higher than in undisturbed regions; the release of nitrogen and organic carbon following logging activities affects nutrient cycling and water quality in forested catchments. The role of timber harvesting in carbon cycling is widely debated. Research suggests that logging practices influence forest carbon storage, with selective harvesting mitigating emissions better than clear-cutting; the use of harvested timber for bioenergy production can partially offset emissions from logging activities; post-harvest regeneration strategies play a crucial role in determining the long-term carbon balance of logged forests.

Sustainable forest management strategies, including selective logging, reduced-impact harvesting, and advanced monitoring techniques, are essential for mitigating these effects while maintaining economic viability.

Future studies should focus on sustainable harvesting methods and their long-term ecological implications. Key takeaways from this review include the importance of adapting harvesting methods to steep terrain, optimizing road networks, and leveraging technologies such as GIS, remote sensing, and machine learning to enhance forest management efficiency.

Additionally, research on the long-term ecological impacts of timber extraction remains limited, particularly regarding the recovery of forest ecosystems following disturbances such as logging, wildfires, and pest outbreaks.

In conclusion, while timber harvesting in mountain forests is necessary for economic and resource management purposes, it must be carefully regulated to ensure long-term ecological sustainability. By integrating advanced technologies, policy frameworks, and sustainable management strategies, it is possible to achieve a balance between economic profitability and environmental conservation, ultimately supporting the resilience of mountain forest ecosystems in the face of climate change and human activities.

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References

1. Mayor, J.R.; Sanders, N.J.; Classen, A.T.; Bardgett, R.D.; Clement, J.C.; Fajardo, A.; Lavorel, S.; Sundqvist, M.K.; Bahn, M.; Chisholm, C.; et al. Elevation alters ecosystem properties across temperate tree lines globally. *Nature* **2017**, *542*, 91–95. [CrossRef] [PubMed]
2. Price, L.W. *Mountains and Man: A Study of Process and Environment*; University of California Press: Berkeley, CA, USA, 1986; p. 271. ISBN 9780520058866.
3. Rundel, P.W.; Parsons, D.J.; Gordon, D.T. Montane and subalpine vegetation of the Sierra Nevada and Cascade Ranges. In *Terrestrial Vegetation of California*; Barbour, M.G., Major, J., Eds.; Wiley: New York, NY, USA, 1977; pp. 559–599.
4. Aparecido, L.M.T.; Teodoro, G.S.; Mosquera, G.; Brum, M.; Barros, F.d.V.; Pompeu, P.V.; Rodas, M.; Lazo, P.; Müller, C.S.; Mulligan, M.; et al. Ecohydrological drivers of Neotropical vegetation in montane ecosystems. *Ecohydrology* **2018**, *11*, e1932. [CrossRef]
5. Dinca, L.; Marin, M.; Radu, V.; Murariu, G.; Drasovean, R.; Cretu, R.; Georgescu, L.; Timiș-Gânsac, V. Which are the best site and stand conditions for silver fir (*Abies alba* Mill.) located in the Carpathian Mountains? *Diversity* **2022**, *14*, 547. [CrossRef]
6. Murariu, G.; Dinca, L.; Tudose, N.; Crisan, V.; Georgescu, L.; Munteanu, D.; Dragu, M.D.; Rosu, B.; Mocanu, G.D. Structural characteristics of the main resinous stands from Southern Carpathians, Romania. *Forests* **2021**, *12*, 1029. [CrossRef]
7. Bürgi, P.; Thomas, M.; Pauli, B.; Auer, N. *Forstwirtschaftliches Testbetriebsnetz der Schweiz: Ergebnisse der Jahre 2014–2016*; Bundesamt für Statistik: Wiesbaden, Germany, 2018; pp. 1–48.
8. Donthu, N.; Kumar, S.; Mukherjee, D.; Pandey, N.; Lim, W.M. How to conduct a bibliometric analysis: An overview and guidelines. *J. Bus. Res.* **2021**, *133*, 285–296. [CrossRef]
9. Baker, H.K.; Kumar, S.; Pattnaik, D. Twenty-five years of Review of Financial Economics: A bibliometric overview. *Rev. Financ. Econ.* **2020**, *38*, 3–23. [CrossRef]
10. Dinca, L.; Crisan, V.; Ienasoiu, G.; Murariu, G.; Drasovean, R. Environmental Indicator Plants in Mountain Forests: A Review. *Plants* **2024**, *13*, 3358. [CrossRef]
11. Palomo, I. Climate change impacts on ecosystem services in high mountain areas: A literature review. *Mt. Res. Dev.* **2017**, *37*, 179–187. [CrossRef]
12. Kräuchi, N.; Brang, P.; Schönenberger, W. Forests of mountainous regions: Gaps in knowledge and research needs. *For. Ecol. Manag.* **2000**, *132*, 73–82. [CrossRef]
13. Mengist, W.; Soromessa, T.; Legese, G. Ecosystem services research in mountainous regions: A systematic literature review on current knowledge and research gaps. *Sci. Total. Environ.* **2020**, *702*, 134581. [CrossRef]

14. Moskalik, T.; Borz, S.A.; Dvořák, J.; Ferencik, M.; Glushkov, S.; Muiste, P.; Lazdiņš, A.; Styranivsky, O. Timber harvesting methods in Eastern European countries: A review. *Croat. J. For. Eng. J. Theory Appl. For. Eng.* **2017**, *38*, 231–241.
15. McEwan, A.; Marchi, E.; Spinelli, R.; Brink, M. Past, present and future of industrial plantation forestry and implication on future timber harvesting technology. *J. For. Res.* **2020**, *31*, 339–351. [CrossRef]
16. Lindroos, O.; La Hera, P.; Häggström, C. Drivers of advances in mechanized timber harvesting–a selective review of technological innovation. *Croat. J. For. Eng. J. Theory Appl. For. Eng.* **2017**, *38*, 243–258.
17. Clarivate. Web of Science Core Collection. 2024. Available online: https://clarivate.com/products/scientific-and-academic-research/research-discovery-and-workflow-solutions/webofscience-platform/web-of-science-core-collection/ (accessed on 5 January 2025).
18. Microfosft. Microsoft Excel. 2024. Available online: https://www.microsoft.com/en-us/microsoft-365/excel?legRedir=true&CorrelationId=3bb60ab0-fe13-41a4-812b-2627667cf346 (accessed on 8 January 2025).
19. Geochart. Available online: https://developers.google.com/chart/interactive/docs/gallery/geochart (accessed on 11 January 2025).
20. VOSviewer. Available online: https://www.vosviewer.com// (accessed on 7 January 2025).
21. Sims, C. Optimal timing of salvage harvest in response to a stochastic infestation. *Nat. Resour. Model.* **2011**, *24*, 383–408. [CrossRef]
22. She, J.; Chung, W.; Han, H. Economic and environmental optimization of the forest supply chain for timber and bioenergy production from beetle-killed forests in northern Colorado. *Forests* **2019**, *10*, 689. [CrossRef]
23. Collins, B.J.; Rhoades, C.C.; Hubbard, R.M.; Battaglia, M.A. Tree regeneration and future stand development after bark beetle infestation and harvesting in Colorado lodgepole pine stands. *For. Ecol. Manag.* **2011**, *261*, 2168–2175. [CrossRef]
24. Dale, V.H.; Joyce, L.A.; McNulty, S.; Neilson, R.P.; Ayres, M.P.; Flannigan, M.D.; Hanson, P.J.; Irland, L.C.; Lugo, A.E.; Peterson, C.J.; et al. Climate change and forest disturbances: Climate change can affect forests by altering the frequency, intensity, duration, and timing of fire, drought, introduced species, insect and pathogen outbreaks, hurricanes, windstorms, ice storms, or landslides. *BioScience* **2001**, *51*, 723–734. [CrossRef]
25. Beschta, R.L.; Rhodes, J.J.; Kauffman, J.B.; Gresswell, R.E.; Minshall, G.W.; Karr, J.R.; Perry, D.A.; Hauer, F.R.; Frissell, C.A. Postfire management on forested public lands of the western United States. *Conserv. Biol.* **2004**, *18*, 957–967. [CrossRef]
26. Donato, D.C.; Fontaine, J.B.; Campbell, J.L.; Robinson, W.D.; Kauffman, J.B.; Law, B.E. Post-wildfire logging hinders regeneration and increases fire risk. *Science* **2006**, *311*, 352. [CrossRef]
27. Greene, D.F.; Gauthier, S.; Noël, J.; Rousseau, M.; Bergeron, Y. A field experiment to determine the effect of post-fire salvage on seedbeds and tree regeneration. *Front. Ecol. Environ.* **2006**, *4*, 69–74. [CrossRef]
28. Rumbaitis del Rio, C.M. Changes in understory composition following catastrophic windthrow and salvage logging in a subalpine forest ecosystem. *Can. J. For. Res.* **2006**, *36*, 2943–2954. [CrossRef]
29. Fraver, S.; Jain, T.; Bradford, J.B.; D'Amato, A.W.; Kastendick, D.; Palik, B.; Shinneman, D.; Stanovick, J. The efficacy of salvage logging in reducing subsequent fire severity in conifer-dominated forests of Minnesota, USA. *Ecol. Appl.* **2011**, *21*, 1895–1901. [CrossRef] [PubMed]
30. Collins, B.; Rhoades, C.; Battaglia, M.; Hubbard, R. The effects of bark beetle outbreaks on forest development, fuel loads and potential fire behavior in salvage logged and untreated lodgepole pine forests. *For. Ecol. Manag.* **2012**, *284*, 260–268. [CrossRef]
31. Chen, H.; Hu, Y.; Chang, Y.; Bu, R.; Li, Y.; Liu, M. Changes of forest fire regime and landscape pattern under different harvesting modes in a boreal forest of Northeast China. *J. Arid. Land* **2015**, *7*, 841–851. [CrossRef]
32. Sterner, S.; Aslan, C.; Best, R.; Chaudhry, T. Forest management effects on vegetation regeneration after a high severity wildfire: A case study in the southern Cascade range. *For. Ecol. Manag.* **2022**, *520*, 120394. [CrossRef]
33. Anderson, G.K. Snow Avalanche Penetration into Mature Forest. Ph.D. Thesis, University of British Columbia, Vancouver, BC, Canada, 2005.
34. Enache, A.; Kühmaier, M.; Visser, R.; Stampfer, K. Forestry operations in the European mountains: A study of current practices and efficiency gaps. *Scand. J. For. Res.* **2016**, *31*, 412–427. [CrossRef]
35. Zabowski, D.; Java, B.; Scherer, G.; Everett, R.; Ottmar, R. Timber harvesting residue treatment: Part 1. Responses of conifer seedlings, soils and microclimate. *For. Ecol. Manag.* **2000**, *126*, 25–34. [CrossRef]
36. Sforza, F.; Ziesak, M.; Lingua, E.; Starke, M. Optimizing tree selection for planning cable yarding operations: A multi-objective modelling approach. *For. Ecol. Manag.* **2025**, *578*, 122489. [CrossRef]
37. Pecora, G.; Moretti, N.; Dias, S.; Póvoa, O. Geographic information systems (GIS) as a tool for the optimization of timber harvesting in mountain regions forests. In *Innovar y Producir Para el Futuro*; Fundacion General Univ Politecnica Madrid: Madrid, Spain, 2014; pp. 1534–1539.
38. Shao, G.; Wang, F.; Dai, L.; Bai, J.; Li, Y. A density-dependent matrix model and its applications in optimizing harvest schemes. *Sci. China Technol. Sci.* **2006**, *49*, 108–117. [CrossRef]

39. Panizzoni, G.; Magliocchetti, D.; Prandi, F.; De Amicis, R. Interactive virtual planning tools for sustainable forest production in mountain areas. In *International Conference on Human-Computer Interaction*; Springer International Publishing: Cham, Switzerland, 2015; pp. 220–225.
40. Bont, L.G.; Fraefel, M.; Frutig, F.; Holm, S.; Ginzler, C.; Fischer, C. Improving forest management by implementing best suitable timber harvesting methods. *J. Environ. Manag.* **2022**, *302*, 114099. [CrossRef]
41. Prandi, F.; Magliocchetti, D.; Poveda, A.; De Amicis, R.; Andreolli, M.; Devigili, F. New Approach for forest inventory estimation and timber harvesting planning in mountain areas: The SLOPE project. *Int. Arch. Photogramm. Remote Sens. Spat. Inf. Sci.* **2016**, *41*, 775–782. [CrossRef]
42. Hernandez, O.; Merritt, R.W.; Wipfli, M.S. Benthic invertebrate community structure is influenced by forest succession after clearcut logging in southeastern Alaska. *Hydrobiologia* **2005**, *533*, 45–59. [CrossRef]
43. Baldigo, B.P.; Murdoch, P.S.; A Burns, D. Stream acidification and mortality of brook trout (*Salvelinus fontinalis*) in response to timber harvest in Catskill Mountain watersheds, New York, USA. *Can. J. Fish. Aquat. Sci.* **2005**, *62*, 1168–1183. [CrossRef]
44. Harper, E.B.; Patrick, D.A.; Gibbs, J.P. Impact of forestry practices at a landscape scale on the dynamics of amphibian populations. *Ecol. Appl.* **2015**, *25*, 2271–2284. [CrossRef]
45. Barry, D.S.; Pauley, T.K.; Maerz, J.C. Amphibian use of man-made pools on clear-cuts in the Allegheny Mountains of West Virginia, USA. *Appl. Herpetol.* **2008**, *5*, 121–128. [CrossRef]
46. Biek, R.; Mills, L.S.; Bury, R.B. Terrestrial and stream amphibians across clearcut-forest interfaces in the Siskiyou Mountains, Oregon. *Northwest Sci.* **2002**, *76*, 129–140.
47. Mahoney, K.R.; Russell, K.R.; Ford, W.M.; Rodrigue, J.L.; Riddle, J.D.; Schuler, T.M.; Adams, M.B. Woodland salamander responses to a shelterwood harvest-prescribed burn silvicultural treatment within Appalachian mixed-oak forests. *For. Ecol. Manag.* **2016**, *359*, 277–285. [CrossRef]
48. Knapp, S.M.; Haas, C.A.; Harpole, D.N.; Kirkpatrick, R.L. Initial effects of clearcutting and alternative silvicultural practices on terrestrial salamander abundance. *Conserv. Biol.* **2003**, *17*, 752–762. [CrossRef]
49. Zurita, G.A.; Zuleta, G.A. Bird use of logging gaps in a subtropical mountain forest: The influence of habitat structure and resource abundance in the Yungas of Argentina. *For. Ecol. Manag.* **2009**, *257*, 271–279. [CrossRef]
50. Weakland, C.A.; Wood, P.B.; Ford, W. Responses of songbirds to diameter-limit cutting in the central Appalachians of West Virginia, USA. *For. Ecol. Manag.* **2002**, *155*, 115–129. [CrossRef]
51. Hayward, G.D. Forest management and conservation of Boreal Owls in North America. *J. Raptor Res.* **1997**, *31*, 4.
52. Clark, D.A.; Anthony, R.G.; Andrews, L.S. Relationship between wildfire, salvage logging, and occupancy of nesting territories by northern spotted owls. *J. Wildl. Manag.* **2013**, *77*, 672–688. [CrossRef]
53. Perry, R.W.; Thill, R.E. Small-mammal responses to pine regeneration treatments in the Ouachita Mountains of Arkansas and Oklahoma, USA. *For. Ecol. Manag.* **2005**, *219*, 81–94. [CrossRef]
54. Horn, B.S.; Babb, G.; Gunther, P.M. Small mammal populations and food selection in relation to timber harvest practices in the western Cascade Mountains. *Northwest Sci.* **1983**, *57*, 32–44.
55. Owen, S.F.; Menzel, M.A.; Edwards, J.W.; Ford, W.M.; Menzel, J.M.; Chapman, B.R.; Wood, P.B.; Miller, K.V. Bat activity in harvested and intact forest stands in the Allegheny Mountains. *North. J. Appl. For.* **2004**, *21*, 154–159. [CrossRef]
56. Castleberry, S.B.; Ford, W.M.; Wood, P.B.; Castleberry, N.L.; Mengak, M.T. Movements of Allegheny woodrats in relation to timber harvesting. *J. Wildl. Manag.* **2001**, *65*, 148–156. [CrossRef]
57. Johnson, T.N.; Buskirk, S.W.; Hayward, G.D.; Raphael, M.G. Timber harvest interacts with broad-scale forest mortality to affect site occupancy dynamics of a vertebrate seed predator. *For. Ecol. Manag.* **2015**, *340*, 95–102. [CrossRef]
58. Incoll, R.; Loyn, R.; Ward, S.; Cunningham, R.; Donnelly, C. The occurrence of gliding possums in old-growth forest patches of mountain ash (Eucalyptus regnans) in the Central Highlands of Victoria. *Biol. Conserv.* **2001**, *98*, 77–88. [CrossRef]
59. Nobert, B.R.; Larsen, T.A.; Pigeon, K.E.; Finnegan, L. Caribou in the cross-fire? Considering terrestrial lichen forage in the face of mountain pine beetle (Dendroctonus ponderosae) expansion. *PLoS ONE* **2020**, *15*, e0232248. [CrossRef]
60. Bobek, B.; Boyce, M.S.; Kosobucka, M. Factors affecting red deer (*Cervus elaphus*) population density in Southeastern Poland. *J. Appl. Ecol.* **1984**, *21*, 881–890. [CrossRef]
61. Parks, P.J.; Bostedt, G.; Kriström, B. An integrated system for management and policy analysis. *Environ. Resour. Econ.* **2002**, *21*, 203–220. [CrossRef]
62. Smith, T.S.; Hardin, P.J.; Flinders, J.T. Response of bighorn sheep to clear-cut logging and prescribed burning. *Wildl. Soc. Bull.* **1999**, *27*, 840–845.
63. Attiwill, P.M. Managing Leadbeater's possum in the mountain ash forests of Victoria, Australia—Reply. *For. Ecol. Manag.* **1995**, *74*, 233–237. [CrossRef]
64. McLean, C.M.; Kavanagh, R.P.; Penman, T.; Bradstock, R. The threatened status of the hollow dependent arboreal marsupial, the Greater Glider (*Petauroides volans*), can be explained by impacts from wildfire and selective logging. *For. Ecol. Manag.* **2018**, *415–416*, 19–25. [CrossRef]

65. Kuzyk, G.W.; Kneteman, J.; Schmiegelow, F.K.A. Winter habitat use by wolves, Canis lupus, in relation to forest harvesting in west-central Alberta. *Can. Field-Nat.* **2004**, *118*, 368. [CrossRef]
66. McNitt, D.C.; Alonso, R.S.; Cherry, M.J.; Fies, M.L.; Kelly, M.J. Influence of forest disturbance on bobcat resource selection in the central Appalachians. *For. Ecol. Manag.* **2020**, *465*, 118066. [CrossRef]
67. Mitchell, M.S.; Powell, R.A. Response of black bears to forest management in the southern Appalachian Mountains. *J. Wildl. Manag.* **2003**, *67*, 692. [CrossRef]
68. Ciarniello, L.M.; Boyce, M.S.; Heard, D.C.; Seip, D.R. Denning behavior and den site selection of grizzly bears along the Parsnip River, British Columbia, Canada. *Ursus* **2005**, *16*, 47–58. [CrossRef]
69. Ezzati, S.; Najafi, A.; Rab, M.; Zenner, E. Recovery of soil bulk density, porosity and rutting from ground skidding over a 20-year period after timber harvesting in Iran. *Silva Fenn.* **2012**, *46*, 521–538. [CrossRef]
70. Santiago, J.; Molinero, J.; Pozo, J. Impact of timber harvesting on litterfall inputs and benthic coarse particulate organic matter (CPOM) storage in a small stream draining a eucalyptus plantation. *For. Ecol. Manag.* **2011**, *262*, 1146–1156. [CrossRef]
71. Dinca, L.; Onet, A.; Samuel, A.D.; Tognetti, R.; Uhl, E.; Bosela, M.; Gömöryová, E.; Bielak, K.; Skrzyszewski, J.; Hukić, E.; et al. Microbial soil biodiversity in beech forests of European mountains. *Can. J. For. Res.* **2021**, *51*, 1833–1845. [CrossRef]
72. Christophel, D.; Höllerl, S.; Prietzel, J.; Steffens, M. Long-term development of soil organic carbon and nitrogen stocks after shelterwood- and clear-cutting in a mountain forest in the Bavarian Limestone Alps. *Eur. J. For. Res.* **2015**, *134*, 623–640. [CrossRef]
73. McNabb, D.; Startsev, A.; Nguyen, H. Soil wetness and traffic level effects on bulk density and air-filled porosity of compacted boreal forest soils. *Soil Sci. Soc. Am. J.* **2001**, *65*, 1238–1247. [CrossRef]
74. Fielding, J.A.H.; Hawks, B.S.; Aust, W.M.; Bolding, M.C.; Barrett, S.M. Estimated erosion from clearcut timber harvests in the southeastern United States. *For. Sci.* **2022**, *68*, 334–342. [CrossRef]
75. Satgada, C.P.; Egusa, T.; Yamakawa, Y.; Imaizumi, F. Forest harvesting affects soil organic carbon and total nitrogen transports by facilitating landslides. *Catena* **2023**, *233*, 107517. [CrossRef]
76. Keith, H.; Lindenmayer, D.; Mackey, B.; Blair, D.; Carter, L.; McBurney, L.; Okada, S.; Konishi-Nagano, T. Managing temperate forests for carbon storage: Impacts of logging versus forest protection on carbon stocks. *Ecosphere* **2014**, *5*, 1–34. [CrossRef]
77. Eliasson, L. Effects of forwarder tyre pressure on rut formation and soil compaction. *Silva Fenn.* **2005**, *39*, 549. [CrossRef]
78. Jewett, K.; Daugharty, D.; Krause, H.H.; Arp, P.A. Watershed responses to clear-cutting: Effects on soil solutions and stream water discharge in central New Brunswick. *Can. J. Soil Sci.* **1995**, *75*, 475–490. [CrossRef]
79. Froehlich, H.A.; McNabb, D.H. Minimizing Soil Compaction in Pacifc Northwest Forests. Paper Presented at Sixth North American Forest Soils Conference On Forest Soils and Treatment Impacts, Knoxville, TN, USA, 15–19 August 1983; Available online: https://forest.eecs.wsu.edu/smp/solo/documents/RPs/Froehlich_McNabb_1983.pdf (accessed on 19 February 2025).
80. Horn, R.; Vossbrink, J.; Becker, S. Modern forestry vehicles and their impacts on soil physical properties. *Soil Tillage Res.* **2004**, *79*, 207–219. [CrossRef]
81. Motha, J.A.; Wallbrink, P.J.; Hairsine, P.B.; Grayson, R.B. Determining the sources of suspended sediment in a forested catchment in southeastern Australia. *Water Resour. Res.* **2003**, *39*, 1056. [CrossRef]
82. Litschert, S.; MacDonald, L. Frequency and characteristics of sediment delivery pathways from forest harvest units to streams. *For. Ecol. Manag.* **2009**, *259*, 143–150. [CrossRef]
83. Stednick, J.D. Monitoring the effects of timber harvest on annual water yield. *J. Hydrol.* **1996**, *176*, 79–95. [CrossRef]
84. Williams, L.R.; Taylor, C.M.; Warren, M.L., Jr.; Clingenpeel, J.A. Large-scale effects of timber harvesting on stream systems in the Ouachita Mountains, Arkansas, USA. *Environ. Manag.* **2002**, *29*, 76–87. [CrossRef] [PubMed]
85. Haschenburger, J.K.; Rice, S.P. Changes in woody debris and bed material texture in a gravel-bed channel. *Geomorphology* **2004**, *60*, 241–267. [CrossRef]
86. Wohl, E.; Scott, D.N. Wood and sediment storage and dynamics in river corridors. *Earth Surf. Process. Landf.* **2016**, *42*, 5–23. [CrossRef]
87. Buffington, J.M.; Lisle, T.E.; Woodsmith, R.D.; Hilton, S. Controls on the size and occurrence of pools in coarse-grained forest rivers. *River Res. Appl.* **2002**, *18*, 507–531. [CrossRef]
88. Wohl, E.; Madsen, S.; MacDonald, L. Characteristics of log and clast bed-steps in step-pool streams of northwestern Montana. *Geomorphology* **1997**, *20*, 1–10. [CrossRef]
89. Cadol, D.; Wohl, E. Variable contribution of wood to the hydraulic resistance of headwater tropical streams. *Water Resour. Res.* **2013**, *49*, 4711–4723. [CrossRef]
90. Dincă, L.; Achim, F. The management of forests situated on fields susceptible to landslides and erosion from the Southern Carpathians. *Sci. Pap. Ser. Manag. Econ. Eng. Agric. Rural. Dev.* **2019**, *19*, 183–188.
91. Anderson, C.J.; Lockaby, B.G. The effectiveness of forestry best management practices for sediment control in the southeastern United States: A Literature Review. *South. J. Appl. For.* **2011**, *35*, 170–177. [CrossRef]
92. Sawyers, B.C.; Bolding, M.C.; Aust, W.M.; Lakel III, W.A. Effectiveness and implementation costs of overland skid trail closure techniques in the Virginia Piedmont. *J. Soil Water Conserv.* **2012**, *67*, 300–310. [CrossRef]

93. Cristan, R.; Aust, W.M.; Bolding, M.C.; Barrett, S.M. Estimated sediment protection efficiences for increasing levels of best management practices on forest harvests in the piedmont, USA. *Forests* **2019**, *10*, 997. [CrossRef]
94. Raison, R.J. A review of the impacts of sustainable harvesting, non-harvest management and wildfire on net carbon emissions from Australian native forests. *Aust. For.* **2024**, *87*, 176–194. [CrossRef]
95. Lee, M.R.; McNeil, D.J., Jr.; Mathis, C.L.; Grozinger, C.M.; Larkin, J.L. Microhabitats created by log landings support abundant flowers and insect pollinators within regenerating mixed-oak stands in the Central Appalachian Mountains. *For. Ecol. Manag.* **2021**, *497*, 119472. [CrossRef]
96. McGarigal, K.; Romme, W.H.; Crist, M.; Roworth, E. Cumulative effects of roads and logging on landscape structure in the San Juan Mountains, Colorado (USA). *Landsc. Ecol.* **2001**, *16*, 327–349. [CrossRef]
97. Campu, V.R.; Borz, S.A. Amount and structure of tree damage when using cut-to-length system. *Environ. Eng. Manag. J.* **2017**, *16*, 2053–2061. [CrossRef]
98. Cântar, I.-C.; Ciontu, C.-I.; Dincă, L.; Borlea, G.F.; Crişan, V.E. Damage and tolerability thresholds for remaining trees after timber harvesting: A case study from southwest Romania. *Diversity* **2022**, *14*, 193. [CrossRef]
99. Kalyashov, V.; Tuan, A.D.; Hitrov, E.; Grigoreva, O.; Gurev, A.; Novgorodov, D. Modern systems of machinery and technologies for timber harvesting and reforestation in mountain forests. *Resour. Technol.* **2022**, *19*, 1–47. [CrossRef]
100. Jodłowski, K.; Kalinowski, M. Current possibilities of mechanized logging in mountain areas. *For. Res. Pap.* **2018**, *79*, 365–375. [CrossRef]
101. Available online: www.newfor.net (accessed on 15 February 2025).
102. Visser, R.; Stampfer, K. Expanding ground-based harvesting onto steep terrain: A review. *Croat. J. For. Eng. J. Theory Appl. For. Eng.* **2015**, *36*, 321–331.
103. Wang, J.; Long, C.; McNeel, J.; Baumgras, J. Productivity and cost of manual felling and cable skidding in central Appalachian hardwood forests. *For. Prod. J.* **2004**, *54*, 45–51.
104. Proto, A.R.; Macrì, G.; Visser, R.; Russo, D.; Zimbalatti, G. Comparison of timber extraction productivity between winch and grapple skidding: A case study in Southern Italian Forests. *Forests* **2018**, *9*, 61. [CrossRef]
105. Mologni, O.; Grigolato, S.; Cavalli, R. Harvesting systems for steep terrain in the Italian Alps: State of the art and future prospects. *Contemp. Eng. Sci.* **2016**, *9*, 1229–1242. [CrossRef]
106. Borz, S.A.; Dinulică, F.; Bîrda, M.; Ignea, G.; Ciobanu, V.D.; Popa, B. Time consumption and productivity of skidding Silver fir (*Abies alba* Mill.) round wood in reduced accessibility conditions: A case study in windthrow salvage logging form Romanian Carpathians. *Ann. For. Res.* **2013**, *56*, 363–375.
107. Russell, F.; Mortimer, D. *A Review of Small-Scale Harvesting Systems in Use Worldwide and Their Potential Application in Irish Forestry*; COFORD: Dublin, Ireland, 2005; p. 56.
108. Moskalik, T. Rozwój technik i technologii maszynowego pozyskiwania drewna [Development of machine timber harvesting techniques and technologies]. *Sylwan* **2002**, *146*, 31–38. (In Polish)
109. Kapral, J. Nowe techniki i technologie leśne przyjazne środowisku [New environmentfriendly forest techniques and technologies]. In *Biblioteczka Leśniczego*, 208th ed; Świat Warszawa: Warszawa, Poland, 2004. (In Polish)
110. Grigolato, S.; Panizza, S.; Pellegrini, M.; Ackerman, P.; Cavalli, R. Light-lift helicopter logging operations in the Italian Alps: A preliminary study based on GNSS and a video camera system. *For. Sci. Technol.* **2016**, *12*, 88–97. [CrossRef]
111. Heinimann, H.R.; Caminada, L. Helicopter logging in Switzerland. Analysis of selective logging operations. In Proceedings of a Joint Symposium of IUFRO 3.06 Forest Operations Under Mountainous Conditions and the 9th Pacific Northwest Skyline Symposium, Campbell River, BC, Canada, 13–16 May 1996.
112. Stampfer, K.; Gridling, H.; Visser, R. Analyses of parameters affecting helicopter timber extraction. *Int. J. For. Eng.* **2002**, *13*, 61–68. [CrossRef]
113. Lindenmayer, D.B.; Franklin, J.F. Managing Stand Structure as Part of Ecologically Sustainable Forest Management in Australian Mountain Ash Forests: Manejo de la Estructura como Parte del Manejo Ecológicamente Sustentable de los Bosques de Fresno de Montaña en Australia. *Conserv. Biol.* **1997**, *11*, 1053–1068. [CrossRef]
114. Seidl, R.; Rammer, W.; Jäger, D.; Currie, W.S.; Lexer, M.J. Assessing trade-offs between carbon sequestration and timber production within a framework of multi-purpose forestry in Austria. *For. Ecol. Manag.* **2007**, *248*, 64–79. [CrossRef]
115. Nordfjell, T.; Bacher, M.; Eriksson, L.; Kadlec, J.; Stampfer, K.; Suadicani, K.; Suwala, M.; Talbot, B. Operational factors influencing the efficiency in conversion. In *Norway Spruce Conversion: Options and Consequences*; Spiecker, H., Hansen, K., Klimo, L., Skovsgaard, J.P., Sterba, H., Von Teuffel, K., Eds.; European Forest Institute: Boston, MA, USA, 2004; pp. 197–223.
116. Luo, X.; He, H.S.; Liang, Y.; Wu, Z. Evaluating simulated effects of succession, fire, and harvest for LANDIS PRO forest landscape model. *Ecol. Model.* **2015**, *297*, 1–10. [CrossRef]
117. Vorovencii, I.; Iordache, E. Identification and analysis of forest disturbances and fragmentation in Giurgeu mountains, Romania, using landsat data. *Rural. Dev.* **2013**, *2013*, 513. [CrossRef]

118. Rahmati, O.; Falah, F.; Naghibi, S.A.; Biggs, T.; Soltani, M.; Deo, R.C.; Cerdà, A.; Mohammadi, F.; Bui, D.T. Land subsidence modelling using tree-based machine learning algorithms. *Sci. Total. Environ.* **2019**, *672*, 239–252. [CrossRef] [PubMed]
119. Shabani, S.; Pourghasemi, H.R.; Blaschke, T. Forest stand susceptibility mapping during harvesting using logistic regression and boosted regression tree machine learning models. *Glob. Ecol. Conserv.* **2020**, *22*, e00974. [CrossRef]
120. Patrick, D.A.; Laxton, C.; Ball, D.; Collins, S.; Korzec, S.; Langevin, C.; Vimislik, J. A multi-scale evaluation of the effects of forest harvesting for woody biofuels on mammalian communities in a northern hardwood forest. *Northeast. Nat.* **2013**, *20*, 678–693. [CrossRef]
121. Kirkland, G.L., Jr. Patterns of initial small mammal community change after clearcutting of temperate North American forests. *Oikos* **1990**, *59*, 313–320. [CrossRef]
122. Brooks, R.T. Abundance, distribution, trends, and ownership patterns of early-successional forests in the northeastern United States. *For. Ecol. Manag.* **2003**, *185*, 65–74. [CrossRef]
123. Hollie, D.R.; George, A.D.; Porneluzi, P.A.; Haslerig, J.M.; Faaborg, J. Avian community response to experimental forest management. *Ecosphere* **2020**, *11*, e03294. [CrossRef]
124. Hunter, W.C.; Buehler, D.A.; Canterbury, R.A.; Confer, J.L.; Hamel, P.B. Conservation of disturbance-dependent birds in eastern North America. *Wildl. Soc. Bull.* **2001**, *29*, 440–455.
125. Buehler, D.A.; Roth, A.M.; Vallender, R.; Will, T.C.; Confer, J.L.; Canterbury, R.A.; Swarthout, S.B.; Rosenberg, K.V.; Bulluck, L.P. Status and conservation priorities of Golden-winged Warbler (*Vermivora chrysoptera*) in North America. *Auk* **2007**, *124*, 1439–1445. [CrossRef]
126. Fox, T.R. Sustained productivity in intensively managed forest plantations. *For. Ecol. Manag.* **2020**, *138*, 187–202. [CrossRef]
127. McNabb, D. Tillage of compacted haul roads and landings in the boreal forests of Alberta, Canada. *For. Ecol. Manag.* **1994**, *66*, 179–194. [CrossRef]
128. Murray, C.; Buttle, J. Infiltration and soil water mixing on forested and harvested slopes during spring snowmelt, Turkey Lakes Watershed, central Ontario. *J. Hydrol.* **2005**, *306*, 1–20. [CrossRef]
129. May, B.; Attiwill, P. Nitrogen-fixation by Acacia dealbata and changes in soil properties 5 years after mechanical disturbance or slash-burning following timber harvest. *For. Ecol. Manag.* **2003**, *181*, 339–355. [CrossRef]
130. Dincă, L.; Murariu, G.; Enescu, C.M.; Achim, F.; Georgescu, L.; Murariu, A.; Timiș-Gânsac, V.; Holonec, L. Productivity differences between southern and northern slopes of Southern Carpathians (Romania) for Norway spruce, silver fir, birch and black alder. *Not. Bot. Horti Agrobot. Cluj-Napoca* **2020**, *48*, 1070–1084. [CrossRef]
131. Douglas, I.; Pietroniro, A. Predicting road erosion rates in selectively logged tropical rain forests. *Int. Assoc. Hydrol. Sci. Publ.* **2003**, *279*, 199–205.
132. Montero, S.A.; Ellis, E.A. Carbon stock recovery from tree regeneration following selective logging in tropical forest of the Yucatan Peninsula, Mexico. *Carbon Manag.* **2023**, *14*, 2238672. [CrossRef]
133. Robles, D.; Boulanger, Y.; Pascual, J.; Danneyrolles, V.; Bergeron, Y.; Drobyshev, I. Timber harvesting was the most important factor driving changes in vegetation composition, as compared to climate and fire regime shifts, in the mixedwood temperate forests of Temiscamingue since AD 1830. *Landsc. Ecol.* **2025**, *40*, 26. [CrossRef]
134. Picchio, R.; Mederski, P.S.; Tavankar, F. How and how much, do harvesting activities affect forest Soil, regeneration and stands? *Curr. For. Rep.* **2020**, *6*, 115–128. [CrossRef]

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